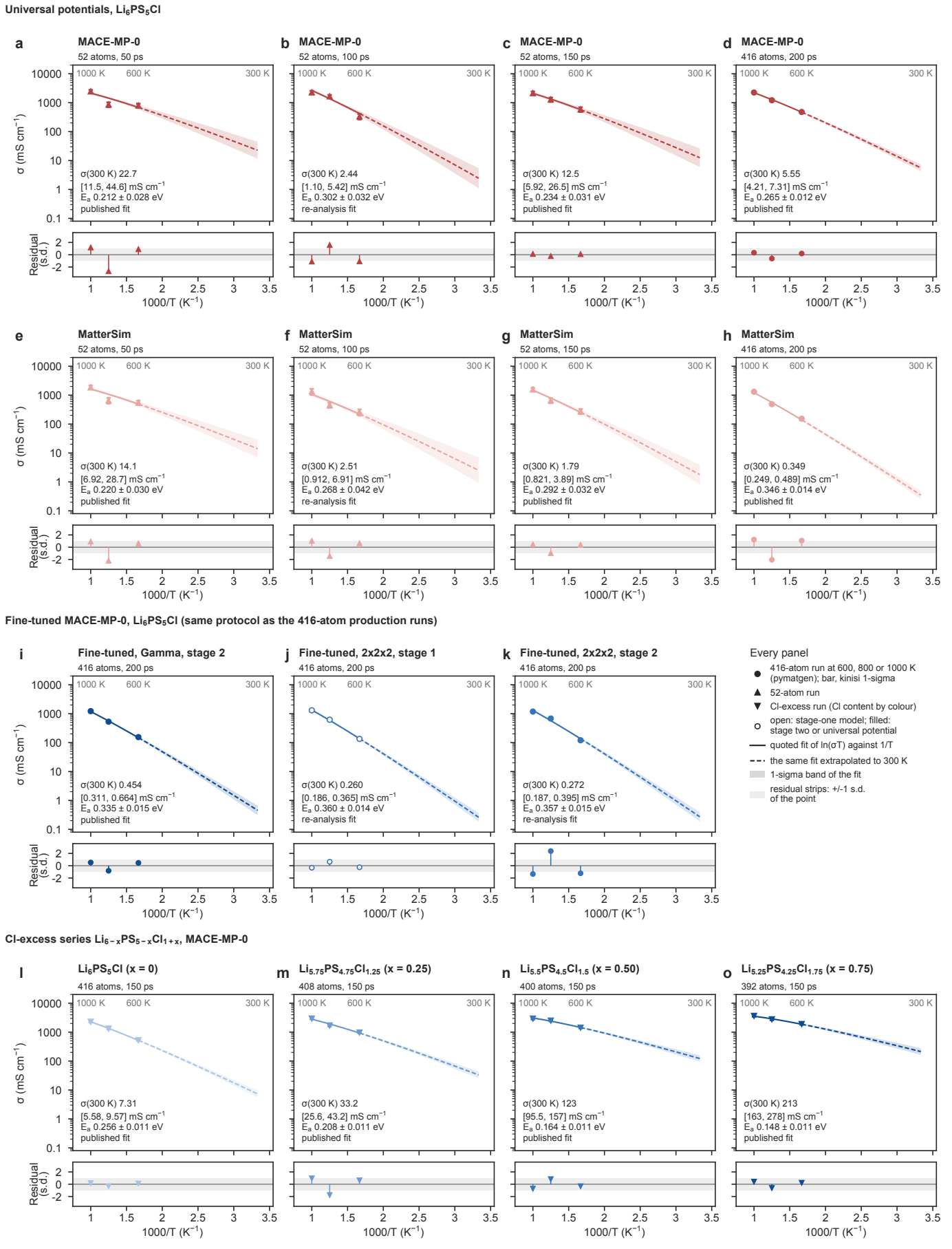

Universal potentials, Li6PS5Cl
a
MACE-MP-0
52 atoms, 50 ps
σ(300 K) 22.7
[11.5, 44.6] mS cm−1
Ea 0.212 ± 0.028 eV
published fit
b
MACE-MP-0
52 atoms, 100 ps
σ(300 K) 2.44
[1.10, 5.42] mS cm−1
Ea 0.302 ± 0.032 eV
re-analysis fit
c
MACE-MP-0
52 atoms, 150 ps
σ(300 K) 12.5
[5.92, 26.5] mS cm−1
Ea 0.234 ± 0.031 eV
published fit
d
MACE-MP-0
416 atoms, 200 ps
σ(300 K) 5.55
[4.21, 7.31] mS cm−1
Ea 0.265 ± 0.012 eV
published fit
e
MatterSim
52 atoms, 50 ps
σ(300 K) 14.1
[6.92, 28.7] mS cm−1
Ea 0.220 ± 0.030 eV
published fit
f
MatterSim
52 atoms, 100 ps
σ(300 K) 2.51
[0.912, 6.91] mS cm−1
Ea 0.268 ± 0.042 eV
re-analysis fit
g
MatterSim
52 atoms, 150 ps
σ(300 K) 1.79
[0.821, 3.89] mS cm−1
Ea 0.292 ± 0.032 eV
published fit
h
MatterSim
416 atoms, 200 ps
σ(300 K) 0.349
[0.249, 0.489] mS cm−1
Ea 0.346 ± 0.014 eV
published fit
1000 K
600 K
300 K
σ (mS cm−1)
Residual (s.d.)
1000/T (K−1)
Fine-tuned MACE-MP-0, Li6PS5Cl (same protocol as the 416-atom production runs)
i
Fine-tuned, Gamma, stage 2
416 atoms, 200 ps
σ(300 K) 0.454
[0.311, 0.664] mS cm−1
Ea 0.335 ± 0.015 eV
published fit
j
Fine-tuned, 2x2x2, stage 1
416 atoms, 200 ps
σ(300 K) 0.260
[0.186, 0.365] mS cm−1
Ea 0.360 ± 0.014 eV
re-analysis fit
k
Fine-tuned, 2x2x2, stage 2
416 atoms, 200 ps
σ(300 K) 0.272
[0.187, 0.395] mS cm−1
Ea 0.357 ± 0.015 eV
re-analysis fit
Every panel
416-atom run at 600, 800 or 1000 K (pymatgen); bar, kinisi 1-sigma
52-atom run
Cl-excess run (Cl content by colour)
open: stage-one model; filled: stage two or universal potential
quoted fit of ln(σT) against 1/T
the same fit extrapolated to 300 K
1-sigma band of the fit
residual strips: +/-1 s.d. of the point
Cl-excess series Li6−xPS5−xCl1+x, MACE-MP-0
l
Li6PS5Cl (x = 0)
416 atoms, 150 ps
σ(300 K) 7.31
[5.58, 9.57] mS cm−1
Ea 0.256 ± 0.011 eV
published fit
m
Li5.75PS4.75Cl1.25 (x = 0.25)
408 atoms, 150 ps
σ(300 K) 33.2
[25.6, 43.2] mS cm−1
Ea 0.208 ± 0.011 eV
published fit
n
Li5.5PS4.5Cl1.5 (x = 0.50)
400 atoms, 150 ps
σ(300 K) 123
[95.5, 157] mS cm−1
Ea 0.164 ± 0.011 eV
published fit
o
Li5.25PS4.25Cl1.75 (x = 0.75)
392 atoms, 150 ps
σ(300 K) 213
[163, 278] mS cm−1
Ea 0.148 ± 0.011 eV
published fit